178064: coarse sandstone, The Stock Well

Location and sampling

YOUANMI (SH 50-4), RICHARDSON (2840)
MGA Zone 50, 792660E 6805480N

Sampled on 31 October 2002

The sample was taken from a 0.5 m-long, 20 cm-thick block 5.5 km northeast of The Stock Well.

Tectonic unit/relations

The sample is a pale pinkish-white, medium- to coarse- and even-grained, well-sorted quartz sandstone from a 20 m-thick, bedded quartzite unit from the lowest exposed stratigraphic unit in the Illaara greenstone belt, Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Chen, 2004). Bedding in the quartzite unit strikes at 350° and dips steeply to the west. The sample was collected to determine the maximum depositional age of the sandstone precursor to the quartzite.

Petrographic description

The principal minerals are quartz (96 vol.%), muscovite (1 vol.%), and aluminosilicate (mostly andalusite, but also sillimanite and kyanite; 2–3 vol.%), with accessory rutile (trace), ?pyrophyllite, ?diaspore or ?both (trace), ?paragonite, margarite or ?both (trace), and zircon (trace). This pale, pinkish quartzite is laminated and has a white weathering rind. Both the mica and aluminosilicate are elongate parallel to the foliation. The mica forms very narrow flakes 0.1 to 1 mm long. The aluminosilicate is mostly andalusite, but rare grains of kyanite and sillimanite are also present, with grains apparently 0.2 to 1 mm long. The quartz occurs as anhedral grains up to 2 mm in diameter, indicating exaggerated grain-growth. There are rare lamellae that are rich in rutile. Zircon, up to 0.1 mm in grain size, is sparse. The grain size of the zircon suggests a former coarse-grained sandstone with minor clay. The aluminosilicates are partly altered to phyllosilicate, possibly pyrophyllite, with or without diaspore. The mineral assemblage suggests metamorphism under conditions close to the aluminosilicate triple point (500–530°C at 400–420 MPa), with subsequent retrogression. Possible micas present include muscovite and paragonite, with margarite less likely.

Zircon morphology

The zircons isolated from this sample are typically colourless, pale yellow, dark brown or black, elliptical or subhedral and with rounded terminations, equant or slightly to substantially elongate, and between 50 × 80 µm and 160 × 80 µm in size. A majority of grains are fractured and have faint internal zonation. Many grains have a speckled appearance in transmitted light, and many are metamict. Grain surfaces are commonly pitted, consistent with detrital transport, and in reflected light, many have irregular, dark patches and zones. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 19 and 26 May 2003. The counter deadtime during both analysis sessions was 24 ns. Twelve analyses of the CZ3 standard were obtained during the first analysis session. Following deletion of the first three standard analyses as outliers, the remaining nine analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 1.62% (1σ). Analyses 1.1 to 10.1 were obtained during the first analysis session. During the second analysis session, nine analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 2.21% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 6.1, 8.1, 24.1, and 30.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Thirty-five analyses were obtained from 35 zircons. Results are given in Table 1, and shown on concordia and Gaussian-summation probability density plots in Figures 2 and 3 respectively.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with at least one recent episode of radiogenic-Pb loss. Nine concordant to highly discordant analyses of nine zircons (2.1, 3.1, 7.1, 11.1, 17.1, 19.1, 23.1, 26.1, 35.1) have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3632 ± 7 Ma (chi-squared = 1.26). The remaining analyses can not be confidently grouped. Concordant analysis 33.1 indicates a $^{207}Pb/^{206}Pb$ date of 3504 ± 8 Ma (± 1σ uncertainty).

The analyses are interpreted to be of detrital zircons. Analysis 33.1 was obtained on a pinkish-brown, 50 × 80 µm rounded, structureless but fractured grain. The date of 3504 ± 8 Ma (± 1σ uncertainty) indicated by the $^{207}Pb/^{206}Pb$ ratio of concordant analysis 33.1 is therefore interpreted as providing a maximum age for deposition of the precursor sandstone.

Potential source rocks within the western part of Australia having ages matching those of the zircons within

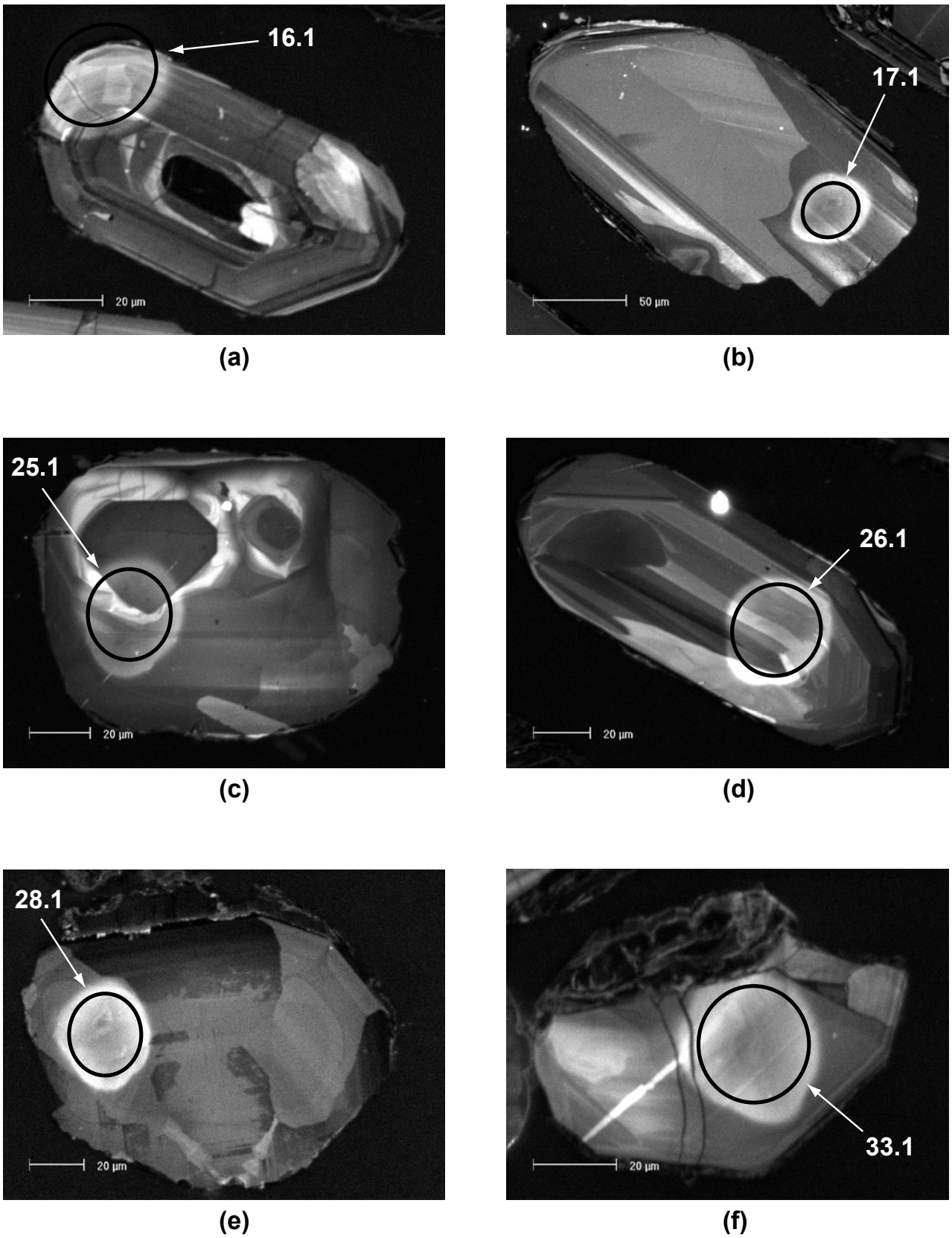

Figure 1. Cathodoluminescence images of representative zircons from sample 178064: coarse sandstone, The Stock Well

Table 1. Ion microprobe analytical results for zircons from sample 178064: coarse sandstone, The Stock Well

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	138	153	153	0.432	0.48834	0.00161	0.14517	0.00160	0.7794	0.0131	52.477	0.921	88	4 206	5
2.1	212	198	202	0.194	0.33532	0.00103	0.18484	0.00122	0.7259	0.0121	33.561	0.581	97	3 642	5
3.1	90	38	85	0.032	0.33587	0.00149	0.11022	0.00139	0.7580	0.0130	35.104	0.640	100	3 644	7
4.1	83	173	42	1.406	0.20469	0.00262	0.01525	0.00508	0.4627	0.0080	13.058	0.297	86	2 864	21
5.1	108	148	51	0.195	0.36044	0.00208	0.03778	0.00224	0.3873	0.0065	19.246	0.355	56	3 752	9
6.1	270	539	174	1.402	0.43684	0.00149	0.21036	0.00235	0.4357	0.0072	26.243	0.454	58	4 041	5
7.1	125	146	95	0.274	0.33060	0.00153	0.12950	0.00198	0.6031	0.0102	27.491	0.496	84	3 620	7
8.1	224	366	170	0.493	0.46360	0.00142	0.20540	0.00167	0.5205	0.0086	33.271	0.572	65	4 129	5
9.1	65	31	73	-0.085	0.40073	0.00190	0.12549	0.00191	0.8680	0.0152	47.960	0.899	103	3 912	7
10.1	114	65	124	0.069	0.36810	0.00131	0.14566	0.00113	0.8319	0.0141	42.222	0.750	103	3 784	5
11.1	175	126	120	0.343	0.33088	0.00181	0.11363	0.00197	0.5469	0.0125	24.953	0.602	78	3 621	8
12.1	111	78	111	0.527	0.32847	0.00211	0.17502	0.00303	0.7604	0.0178	34.438	0.860	101	3 610	10
13.1	167	428	102	0.606	0.43972	0.00222	0.13417	0.00259	0.4436	0.0101	26.895	0.641	58	4 051	8
14.1	62	69	71	0.453	0.38323	0.00272	0.28438	0.00366	0.7884	0.0191	41.660	1.087	97	3 845	11
15.1	142	73	129	0.406	0.36269	0.00183	0.13484	0.00200	0.6950	0.0160	34.755	0.838	90	3 761	8
16.1	186	349	173	0.998	0.35169	0.00189	0.16810	0.00279	0.6876	0.0157	33.343	0.803	91	3 714	8
17.1	142	144	147	0.180	0.33103	0.00156	0.26221	0.00197	0.7528	0.0173	34.360	0.826	100	3 622	7
18.1	98	39	84	0.414	0.33906	0.00213	0.06462	0.00251	0.6978	0.0163	32.622	0.814	93	3 659	10
19.1	124	136	108	0.375	0.33463	0.00190	0.14516	0.00235	0.6742	0.0156	31.108	0.761	91	3 638	9
20.1	180	101	167	0.381	0.33838	0.00153	0.10840	0.00177	0.7366	0.0168	34.368	0.818	97	3 655	7
21.1	213	312	117	0.710	0.38718	0.00216	0.08824	0.00247	0.4254	0.0097	22.710	0.549	59	3 860	8
22.1	154	128	154	0.028	0.36828	0.00159	0.18567	0.00181	0.7494	0.0171	38.054	0.905	95	3 784	7
23.1	146	198	123	0.502	0.33206	0.00187	0.18456	0.00244	0.6353	0.0145	29.088	0.705	87	3 627	9
24.1	265	367	227	0.475	0.36532	0.00108	0.19042	0.00137	0.6276	0.0140	31.611	0.724	83	3 772	4
25.1	96	51	107	0.370	0.40208	0.00166	0.13363	0.00181	0.8333	0.0190	46.196	1.094	100	3 917	6
26.1	113	113	114	0.140	0.33317	0.00144	0.25782	0.00183	0.7375	0.0167	33.876	0.800	98	3 632	7
27.1	157	165	120	0.650	0.35719	0.00153	0.08530	0.00189	0.6064	0.0137	29.864	0.702	82	3 738	7
28.1	163	65	164	0.097	0.36443	0.00125	0.10285	0.00106	0.7976	0.0181	40.076	0.934	100	3 768	5
29.1	126	197	85	1.217	0.36142	0.00225	0.12754	0.00338	0.5053	0.0116	25.179	0.615	70	3 756	9
30.1	236	348	168	0.871	0.33803	0.00146	0.05042	0.00207	0.5762	0.0129	26.855	0.628	80	3 654	7
31.1	121	93	134	0.183	0.47646	0.00188	0.14147	0.00151	0.7925	0.0182	52.060	1.237	90	4 170	6
32.1	161	201	165	0.386	0.52900	0.00198	0.15635	0.00182	0.7007	0.0158	51.111	1.195	79	4 324	5
33.1	136	121	130	0.289	0.30661	0.00170	0.22775	0.00229	0.7193	0.0166	30.408	0.743	100	3 504	9
34.1	34	18	35	0.996	0.34945	0.00429	0.12918	0.00646	0.7717	0.0200	37.182	1.117	100	3 705	19
35.1	92	121	67	0.742	0.33196	0.00263	0.11336	0.00370	0.5732	0.0134	26.237	0.674	81	3 626	12

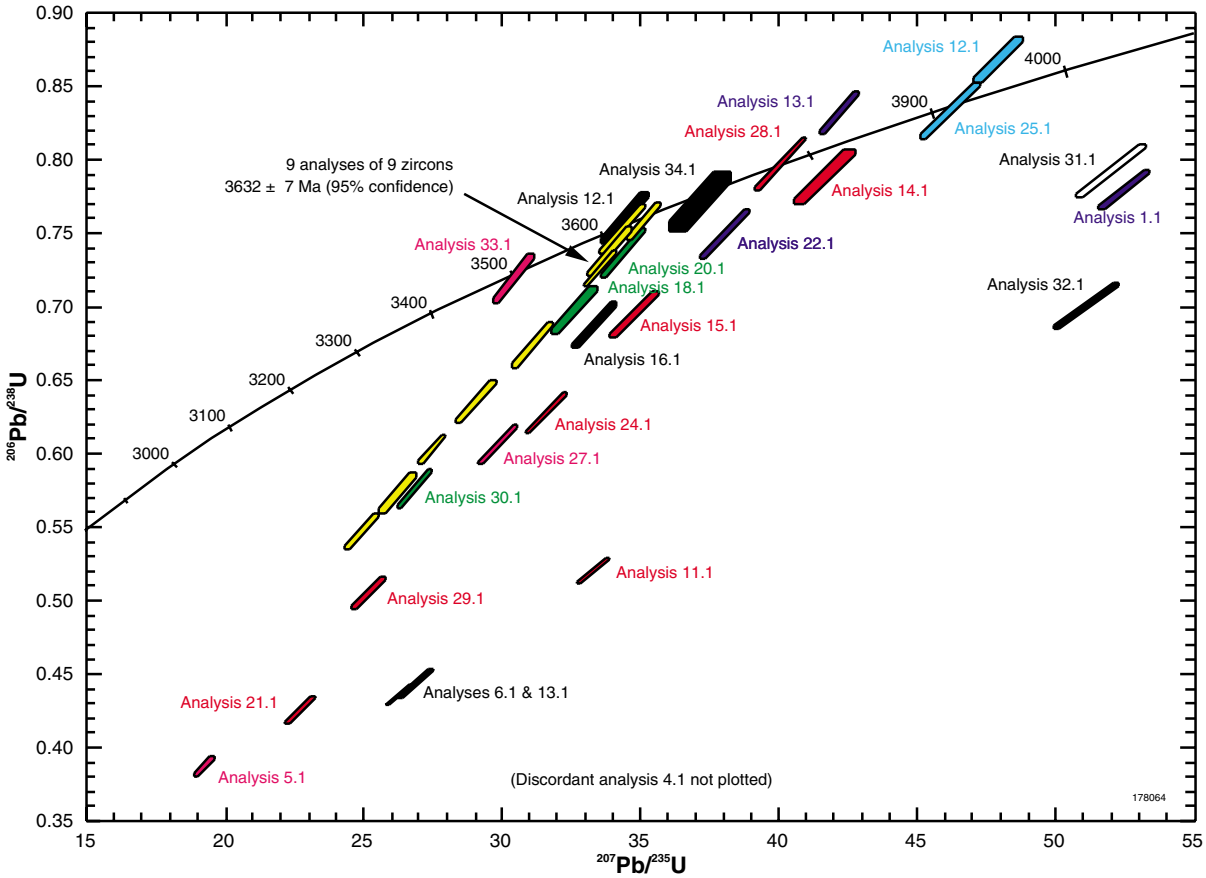

Figure 2. Concordia plot for sample 178064: coarse sandstone, The Stock Well

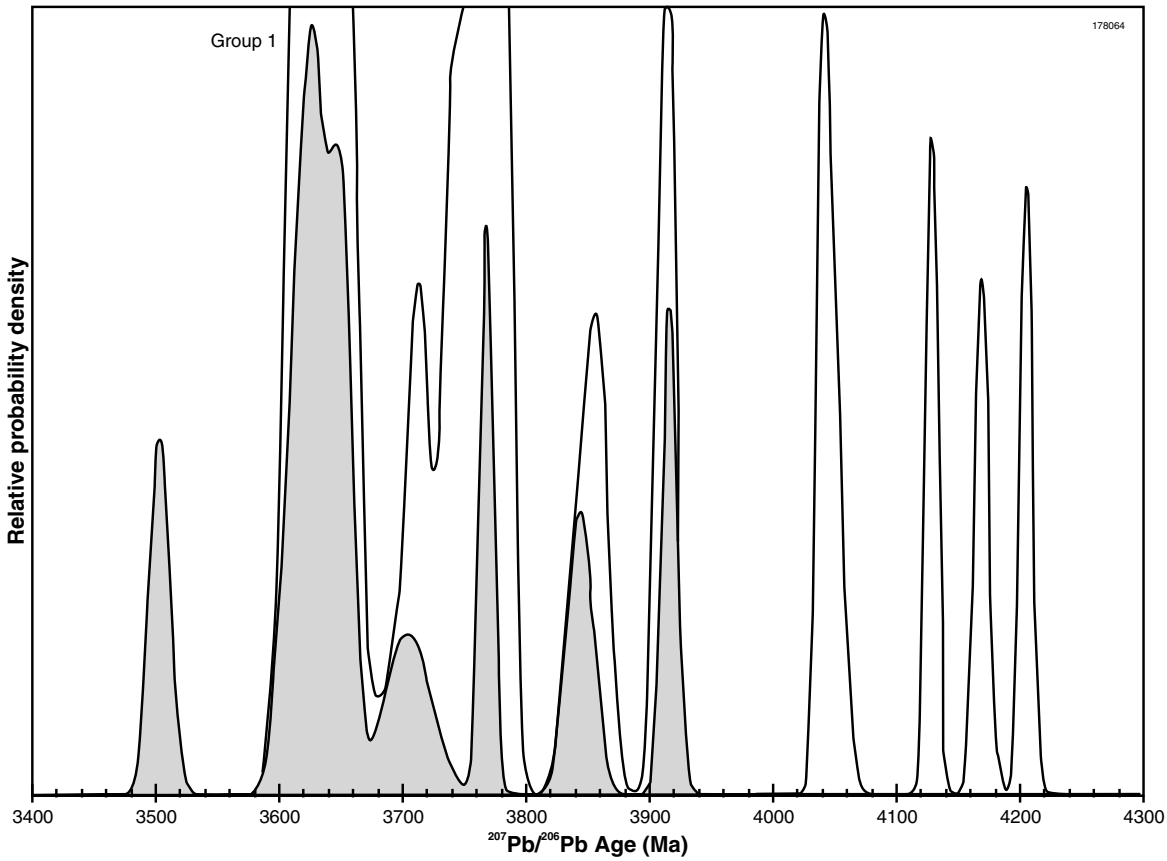

Figure 3. Gaussian-summation probability density plot for sample 178064: coarse sandstone, The Stock Well

this sample include those of the Narryer Terrane in the northwestern part of the Yilgarn Craton (Nelson, 1995, 1996, 1997) and the eastern part of the Pilbara Craton (Nelson, 1996, 1997, 1998, 1999, 2000, 2001, 2002, this compilation).

Recommended reference for this publication:

NELSON, D. R., 2005, 178064: coarse sandstone, The Stock Well; Geochronology dataset 591; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 26/05/2003; Data released: 30/06/2005